

“Garbage to Electricity Converter”

A

Mini Project Report

BACHELOR OF TECHNOLOGY

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CERTIFICATE

Certified that **Akash Baghel , Arpan Saraswat , Shiva Gupta** and **Ritik Rathore** students of Electrical and Electronics Engineering has carried out the project work entitled **“GARBAGE TO ELECTRICITY CONVERTER”** for the Bachelor of Technology in Electrical & Electronics Engineering from Dr. A.P.J Abdul Kalam Technical University, Lucknow. The Mini Project embodies results of original work and studies as are carried out by the students themselves and the content of this project has not been submitted for the award of any other degree/diploma to any other University/Institution.

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ABSTRACT

Electricity generation from waste materials (like paper, dry leaves, etc), also known as thermal waste-to-energy, is a process that involves converting waste materials into electricity by burning them in a combustion chamber. This process is a sustainable solution for waste management as it reduces the volume of waste sent to landfills while producing renewable energy. The methodology for electricity generation by burning waste material typically involves waste collection, handling, and preparation, incineration, energy recovery, and ash management. The generated electricity can be used to power local communities and industries. The process of electricity generation by waste material provides a reliable source of electricity while reducing greenhouse gas emissions by avoiding the release of methane gas from landfills.

Overall, electricity generation by burning waste materials is a promising solution for waste management and renewable energy production. However, it is important to carefully consider the potential benefits and drawbacks of this method and to ensure that appropriate regulation and technologies are in place to minimize any negative environmental impact.

Keywords – Heating panels, Led Bulbs, Jar box, IN4007, Battery 4.5V, Resistors, and Capacitor

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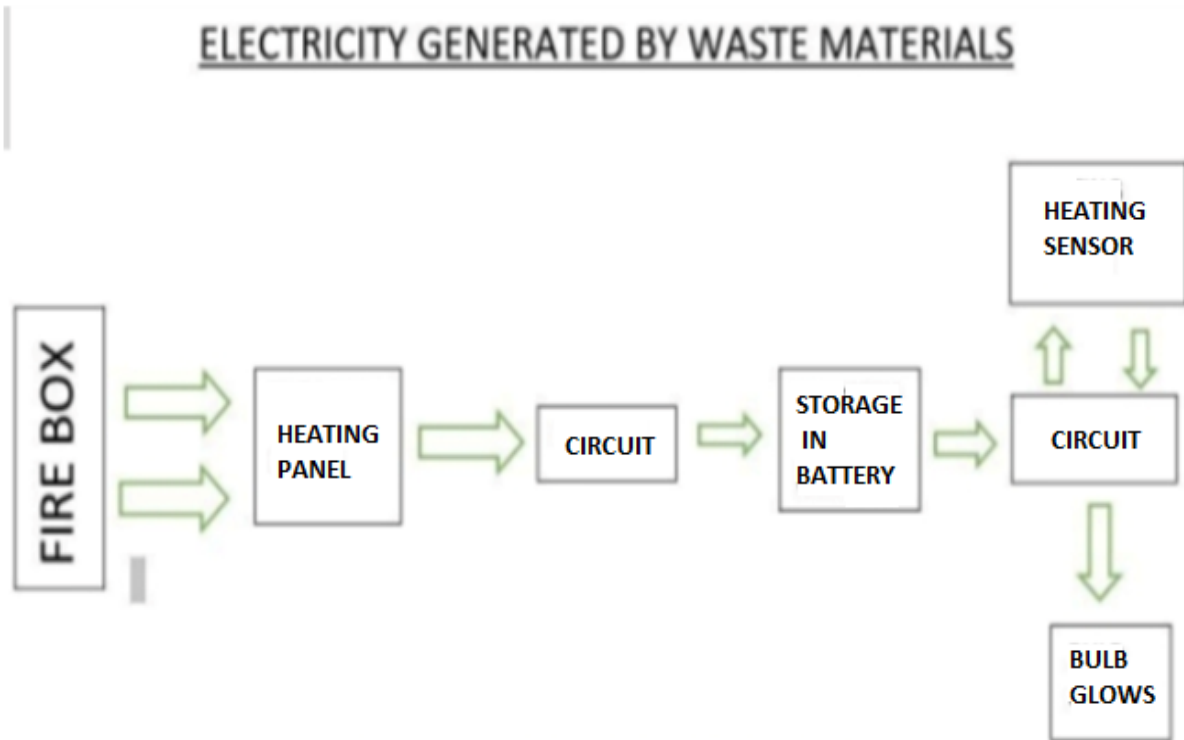
Chapter-1

INTRODUCTION

Electricity generation from waste material is a rapidly growing field that involves the conversion of various types of waste into usable energy. This process is often referred to as waste-to-energy, and it offers a number of benefits including reducing waste in landfills, reducing greenhouse gas emissions, and providing a source of renewable energy. The process of generating electricity from waste materials typically involves the use of thermal or biological processes. Thermal processes involve the incineration of waste, which is then used to generate steam to power turbines and produce electricity. One of the main advantages of electricity generation by burning waste materials is that it reduces the volume of waste sent to landfills, which in turn reduces the amount of space required for landfill sites. This can help to mitigate the negative impacts of landfill sites on the environment, such as groundwater contamination and greenhouse gas emissions. Another advantage of electricity generation by burning waste materials is that it produces a reliable source of electricity. This is particularly important in areas where there may be limited access to other sources of electricity, such as remote communities or developing countries. The technology used to generate electricity from these waste materials is constantly evolving, and new innovations are being developed to make the process more efficient and cost-effective. Overall, the generation of electricity from waste material represents an exciting opportunity to reduce waste, lower greenhouse gas emissions, and provide a source **of renewable energy for communities around the world.**

Chapter-2

BLOCK DIAGRAM



In this Block Diagram you can see when we burn waste materials and fire box then heat generating and heating panel starts to heat convert electricity and after that that electricity we can see by LED Bulb glowing and that electricity go to circuit and after that in battery and start storing power and when electricity store in battery then heating sensor turn on the output power supply and LED Bulb start glowing and smoke go to water tank and filter system start controlling pollution.

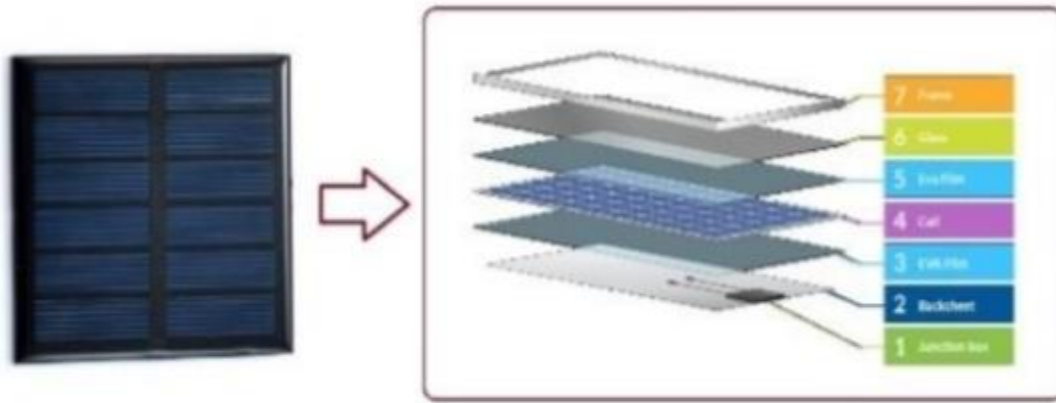
MODEL DIAGRAM

(GARBAGE TO ELECTRICITY CONVERTER)



Chapter-3(COMPONENT USED)

HEATING PANEL (5V)



We used three heating panel , each heating panel is of 5V Heating panels can be used in the process of electricity generation from waste material in several ways, depending on the specific technology used. For example, in thermal waste-to-energy plants, waste material is burned in a combustion chamber to produce high-temperature gases. These gases are then used to heat water and produce steam, which drives a turbine to generate electricity. In this process, heating panels may be used to line the combustion chamber, **as well as other areas of the plant where heat is generated or transferred.** The heating panels can help to maintain the high temperatures required for efficient combustion and steam production, while also protecting the plant's infrastructure from heat damage.

HEATING SENSOR

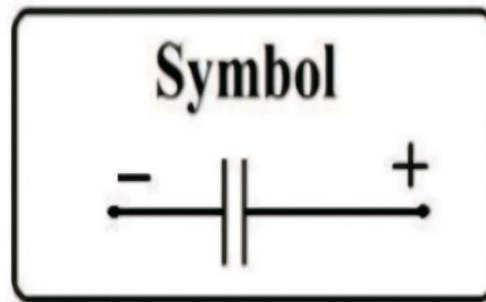


We have used one heating sensor. Heating sensor fitted front of jar box at bottom its connected with the capacitor through a wire Heating sensors are an important component of many waste-to-energy technologies that involve thermal processes, such as incineration and gasification. These sensors are used to measure temperature in various parts of the process, providing valuable information that can be used to optimize the efficiency of the system and prevent damage.

CAPACITOR

Electrolytic Capacitor

4700uf/35v



We have used 18 capacitor Capacitors can be used in various ways in the process of electricity generation from waste material. Capacitors are electrical components that store and release electrical energy, and they are commonly used in electrical systems to provide power factor correction, voltage regulation, and energy storage. In waste-to-energy plants that use thermal processes such as incineration, capacitors may be used in the electrical systems that control the turbines and generators that produce electricity. These capacitors can help to regulate the voltage and power factor of the electrical output, ensuring that the electricity produced is stable and reliable.

RESISTOR



resistor



resistor symbol

Resistors can be used in various ways in the process of electricity generation from waste material. Resistors are electrical components that resist the flow of current in a circuit, and they are commonly used in electrical systems to control current, voltage, and power. In waste-to-energy plants that use thermal processes such as incineration, resistors may be used in the electrical control systems that regulate the temperature and flow of gases in the combustion chamber. These resistors can help to control the flow of current to heating elements or other devices that generate heat, ensuring that the temperature is maintained within the desired range for efficient combustion.

LITHIUM BATTERY(3.7V)

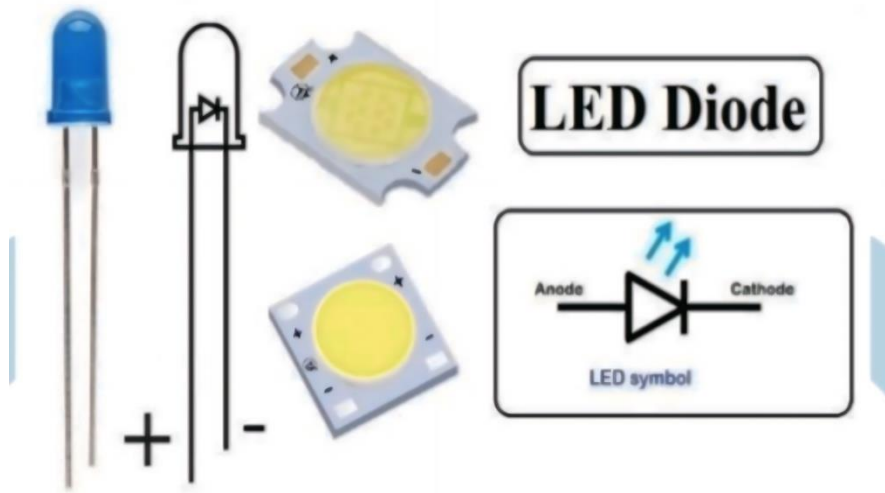


THIS LITHIUM BATTERY IS RECHARGABLE

We have used four lithium battery one is connected with the heating panel and three battery connected with the DC fan. The battery could serve as a backup power source in case the converter's output drops or fails. This ensures that the connected devices, such as the light bulbs, remain powered even if the converter experiences issues.

Power the Fan: Connect the three lithium batteries in parallel to power the fan. This will create a higher voltage, allowing the fan to operate efficiently

(DC)LED BULB(7W)



We have used Three DC LED in this project. LED bulbs can be used in various ways in the process of electricity generation from waste material. LED bulbs are a type of lighting technology that use light-emitting diodes (LEDs) to produce light, and they are known for their energy efficiency, long lifespan, and durability. In waste-to-energy plants, LED bulbs may be used to provide lighting in various areas, including control rooms, storage areas, and maintenance facilities. LED bulbs consume less energy than traditional incandescent bulbs, and they have a longer lifespan, which can help to reduce energy consumption and maintenance costs over time.

MULTIMETER



Voltage Measurement: It measures the voltage output of the converter system. This helps determine the efficiency of the energy conversion process. By monitoring the voltage, you can assess whether the system is generating sufficient power to operate the connected devices.

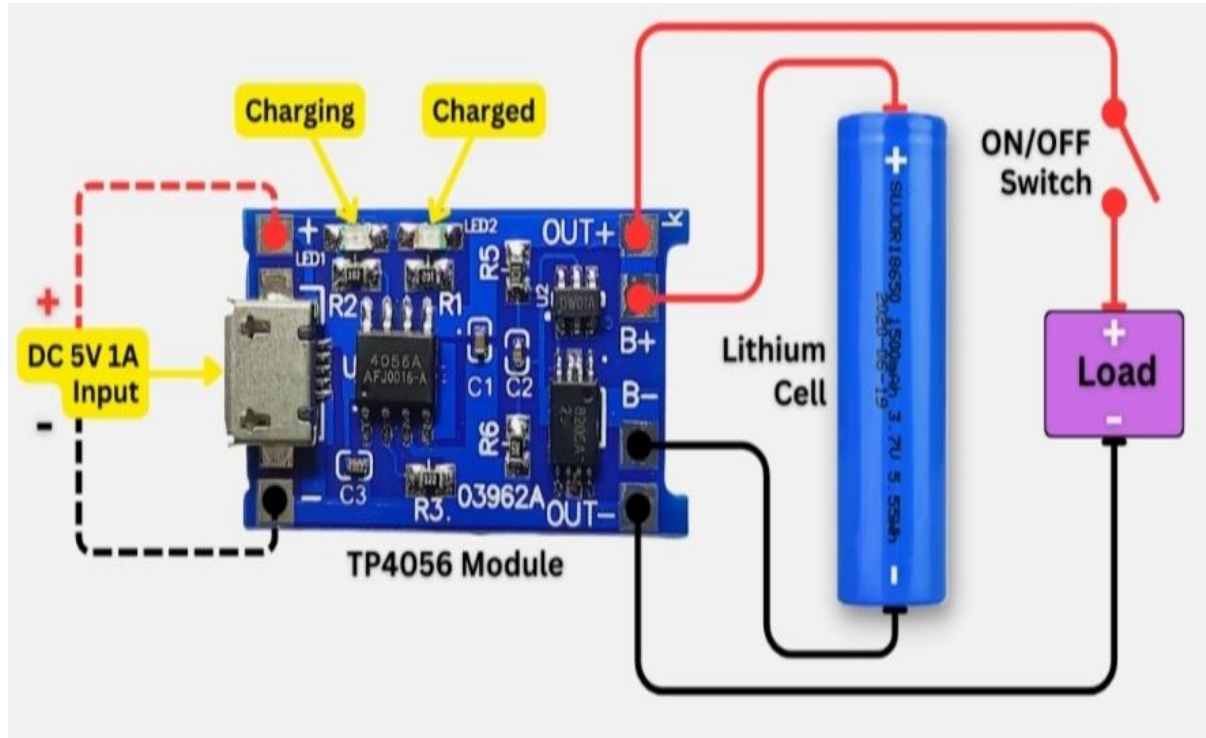
Connect the anode and cathode to the multimeter. The multimeter will measure the voltage and current generated by the garbage from solar panels.

DC FAN (12V)



The fan, powered by the lithium batteries, draws air. This helps to remove carbon dioxide from the waste, improving the efficiency. The captured carbon dioxide can be stored or used for other purposes, such as carbon sequestration. This fan is fitted on a plastic box and a wet foam in this box on which all the carbon collects.

TP4056 CHARGING MODULE



The TP4056 is a complete constant current/constant-voltage linear charger for single cell lithium-ion batteries. Its SOP package and low external component count make the TP4056 ideally suited for portable applications. Furthermore, the TP4056 can work within USB and wall adapter.

Chapter-5

METHODOLOGY

When we start burning the waste material in the burning box the heating panels will start collecting the generated heat energy by waste material in the burning box. The heat energy collected by heating panel will be converted into the electrical energy. The generated electrical energy will be seen in circuit box with led glowing. The generated electrical energy will transfer to the batteries through the power boosters. The batteries will not dissipate the energy back because a diode is connected to it. The batteries relate to the heat sensor and LED bulbs. Whenever the heat sensor will start conducting the batteries allow energy to flow will start conducting and LED bulbs will glow.

RESULT

In this prototype when we start heating waste material inside jar box the heat generated will be collected by heating panels. The heating panels will collect the converted energy (heat energy into the electrical energy) which will be transferred to circuit board. Which is constructed with IN4007 Diode and capacitor connected in series and parallel because in to increase the generated energy and push to store in the battery. Then the heating sensor is will be sensing the heat and connect circuit to output of LED Bulbs and the bulbs will glow until the energy getting stored and until the heating sensor sensing the energy generation. The bulb will glow uninterruptedly while the energy generation and battery storage. This stored energy can be used for anything. In the present situation waste material at any place, we can see with some procedure collect everything and we can use the prototype to generate the more energy for utilization. With this we came know that the energy generation by this method is very easy with some precaution. This prototype helps us to know about waste utilization. With this project we can increase our own energy at industrial purpose and use them for some needs.

Chapter-6

CONCLUSION

Incineration technology is complete combustion of waste (Municipal Solid Waste or Refuse derived fuel) with the recovery of heat to produce energy that in turn produces power through heating panels. Now from this we can conclude that electricity plays an important role in our life we are made aware of how the generate electricity waste is done. For technical service provider plant Objectives & Maintenance activities are very important as its service mostly depends on the availability of its equipment. From this we see that how electricity generated successfully. From this we can see how to store the energy in batteries.

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